Unit 2

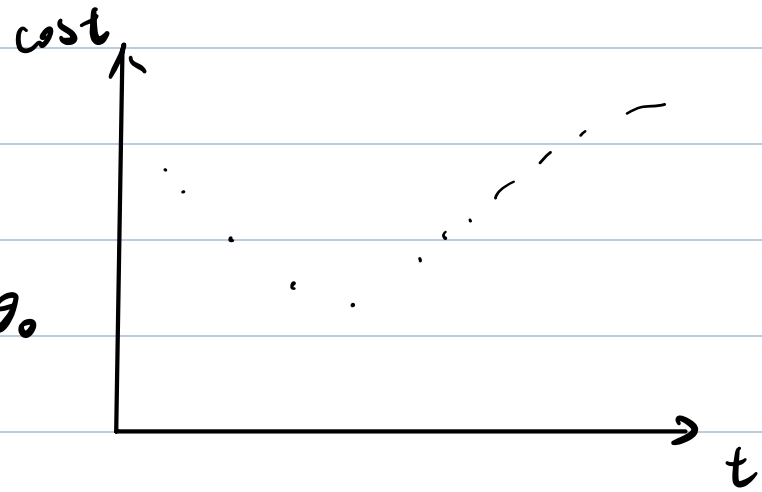
Classification

$$S_n = \{ (x^{(t)}, y^{(t)}) \mid t = 1, \dots, n \}$$

$$x^{(t)} = \mathbb{R}^d \quad , y^t \in \{-1, +1\}$$

Linear Regression

$y \in \mathbb{R}$ (continuous value)



Linear

$$f(x, \theta, \theta_0) = \sum_{t=1}^{d} \theta_i x_i + \theta_0$$

$$= \theta x + \theta_0$$

Emperical Risk

$$R_n(\theta) = \frac{1}{n} \sum_{t=1}^{n} \frac{(y^{(t)} - \theta x^{(t)})^2}{2}$$

$$R_{n'}(\theta) = \frac{1}{n'} \sum_{t=n+1}^{n+n'} \frac{(y^{(t)} - \theta x^{(t)})^2}{2}$$

structural mistakes : Training vectors and y

is not linearly.

Estimation mistake: not enough training data

Gradient-based Approach

$$\nabla_\theta (y^{(t)} - \theta \cdot x^{(t)})^2/2 = -(y^{(t)} - \theta \cdot x^{(t)}) \cdot x^{(t)}$$

Although initialize $\theta = 0$

randomly pick $t = \{1 \ldots\ldots n\}$

$$\theta = \theta + \eta((y^{(t)} - \theta \cdot x^{(t)}) \cdot x^{(t)})$$

$$\eta_k = \frac{1}{1+k}$$

Closed Form Solution

$$R_n(\theta) = \frac{1}{n} \sum_{t=1}^{n} (y^{(t)} - \theta \cdot x^{(t)})^2/2$$

$$\frac{dR_n(\theta)}{d\theta} = -\frac{1}{n} \sum_{t=1}^{n} x^{(t)} \cdot (y^{(t)} - \theta \cdot x^{(t)}) = -\frac{1}{n} \sum_{t=1}^{n} y^{(t)} \cdot x^{(t)} + \frac{1}{n} \sum_{t=1}^{n} \theta \cdot (x^{(t)})^2$$

$$= -b + \frac{1}{n} \sum_{t=1}^{n} (x)^{(t)} \cdot \theta \cdot x^{(t)} = -b + \frac{1}{n} \left(\sum_{t=1}^{n} x^{(t)} \cdot (x^{(t)})^T\right) \cdot \theta$$

$$= -b + A \cdot \theta = 0$$

$$\Rightarrow A\theta = b$$

$$\theta = A^{-1} \cdot b$$

$$A = \frac{1}{n} \left(\sum_{t=1}^{n} x^{(t)} \cdot (x^{(t)})^T\right), \quad b = \frac{1}{n} \sum_{t=1}^{n} y^{(t)} \cdot x^{(t)}$$

$$R_n(\theta) = \frac{1}{n}\sum_{t=1}^{n}(y^{(t)} - \theta\cdot x^{(t)})^2/2 + \frac{1}{2}\cdot\lambda\cdot\theta^2$$

$$R_n'(\theta) = \frac{1}{n}\sum_{t=1}^{n} -x^{(t)}(y^{(t)} - \theta\cdot x^{(t)}) + \lambda\cdot\theta$$

$$= -\frac{1}{n}\sum_{t=1}^{n} x^{(t)}\cdot y^{(t)} + \theta\cdot\frac{1}{n}\sum_{t=1}^{n} x^{(t)}\cdot(x^{(t)})^T + \lambda\cdot\theta$$

$$= \theta\cdot\left(\lambda I + \frac{1}{n}\cdot\sum_{t=1}^{n} x^{(t)}\cdot(x^{(t)})^T\right) - \frac{1}{n}\cdot\sum_{t=1}^{n} x^{(t)}\cdot(y^{(t)})^T$$

$$R_n'(\theta) = 0$$

$$\theta\cdot\left(\lambda I + \frac{1}{n}\cdot\sum_{t=1}^{n}(x^{(t)})^T\cdot x^{(t)}\right) = \frac{1}{n}\sum_{t=1}^{n} x^{(t)}\cdot y^{(t)}$$

$$\theta = (X^T\cdot X + \lambda I)^{-1}\cdot X^T Y$$

X = $n\times p$ 的矩阵

Y = R^n 的向量

Regularization

$$J_{\lambda,n}(\theta) = \frac{\lambda}{2}|\theta|^2 + R_n(\theta) = \frac{\lambda}{2}|\theta|^2 + \frac{1}{n}\sum_{t=1}^{n}(y^{(t)} - \theta\cdot x^{(t)})^2/2$$

$$\nabla\theta\left(\frac{1}{2}\lambda|\theta|^2 + \frac{1}{n}\sum_{t=1}^{n}(y^{(t)} - \theta\cdot x^{(t)})^2/2\right) = \lambda\theta - (y^{(t)} - \theta\cdot x^{(t)})\cdot x^{(t)}$$

initialize $\theta = 0$

randomly pick $t = \{1 \ldots n\}$

$$\theta = \theta - \eta\cdot(\lambda\theta - (y^{(t)} - \theta\cdot x^{(t)})\cdot x^{(t)})$$

$$= (1-\eta\lambda)\theta + \eta(y^{(t)} - \theta x^{(t)})\cdot x^{(t)}$$